



UNIVERSITY of
GREENWICH



WEB PROGRAMMING TECHNOLOGIES 1

WT1 COURSEWORK



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Executive Summary

Overview of project

By offering a cooperative environment for knowledge sharing and problem-solving, the Student Learning Adventure Platform is a cutting-edge web application created to improve peer-to-peer learning and student engagement. The software creates a helpful learning community by allowing users to publish, manage, and reply to queries about coursework. This system was created in accordance with the goals and learning objectives specified for the Website Technologies (COMP1841) course. It acts as a test version of a simplified, scalable solution that complies with web development standards and promotes knowledge sharing.

Overallly

The Student Learning Adventure Platform is an online tool designed to foster academic collaboration and enhance peer-to-peer learning. It features a public forum where students can interact by adding, editing, and removing questions, with the option to attach images for better clarity. An interactive admin panel allows administrators to manage users, posts, and academic modules, while a contact form provides direct communication between students and administrators. The platform also organizes questions by academic subject to improve accessibility. With a scalable design, it has the potential for future expansions, including advanced analytics and gamification.

Developed using modern technologies like MySQL, PHP PDO, HTML, CSS, and JavaScript, the platform offers a responsive, user-friendly interface. XAMPP is used for local testing. The project aligns with learning objectives in web application development, critical evaluation of internet technologies, and addressing challenges related to scalability and usability. Overall, the platform promotes academic success through usability, accessibility, and collaboration, fostering a culture of shared learning and continuous improvement.

Methodology, Findings and conclusions

The Student Learning Adventure Platform was designed to enhance academic collaboration through features like a public forum, modular categorization, and an admin panel for easy management. Built with PHP PDO, MySQL, HTML, CSS, and JavaScript, it ensures secure database functionality, a responsive design, and user-friendly access. Developed and tested using XAMPP, the platform supports CRUD operations, image uploads, and admin tools. While it successfully facilitates collaborative learning, future improvements could focus on scalability and security. Overall, the platform serves its purpose well and provides a strong foundation for future growth.

Proposed Specification and Scope (Functional and technical requirements.)

CATEGORY (FUNCTIONAL REQUIREMENTS)	SPECIFICATION
Public Forum for Questions	Students can view, post, edit, and delete questions; images can be attached to questions for clarity.
Admin Panel	Admins can manage users, posts, and assign questions to academic modules.
Contact Form	Students can send messages or queries directly to administrators.
Scalability features	The platform is designed to support future integrations such as gamified learning, analytics, and user tracking.

CATEGORY (TECHNICAL REQUIREMENTS)	SPECIFICATION
Development Environment	XAMPP used for local development, testing, and database management.
Back-End Development	PHP PDO for secure and efficient database interaction; MySQL for relational data management.
Front-End Development	HTML & CSS for responsive design; JavaScript for interactivity and dynamic functionality.
Database Design	Relational schema for structured data storage; PDO and prepared statements for security.
Accessibility and Usability	Supports major web browsers and optimized for different screen sizes and devices.
Legal and Compliance	Adheres to GDPR for data privacy and secure handling of user information.

CATEGORY (PROJECT SCOPE)	SPECIFICATION
Platform Scope	Central hub for academic collaboration, enabling students to share and engage with educational content.
Admin Responsibilities	Admins oversee platform content, manage user roles, and maintain the integrity of academic modules.
Scalability and Future Plans	Supports future features like gamification and advanced user analytics.

Technologies Used in Development

JavaScript, MySQL, PHP PDO, HTML, CSS, XAMPP and Browsers

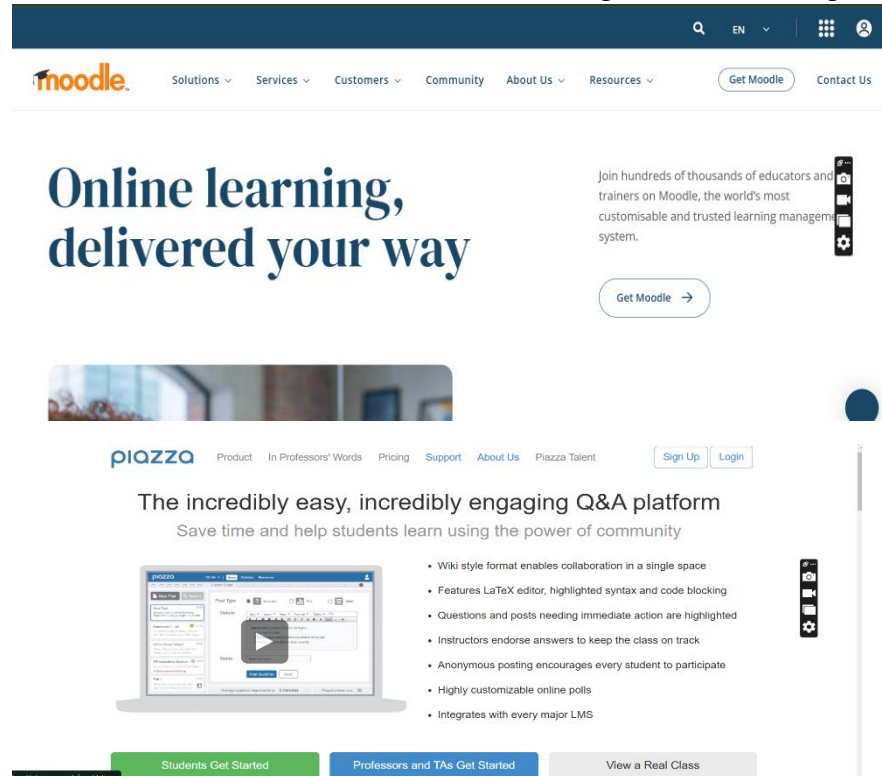
The Student Learning Adventure Platform is built using a mix of modern technologies to make it secure, user-friendly, and interactive. PHP PDO is used to safely communicate with the MySQL database, ensuring user data is securely managed. The platform's layout is designed using HTML5 and CSS, which makes it responsive and adaptable to different devices and screen sizes. For a more dynamic experience, JavaScript and AJAX are implemented, allowing real-time updates without the need to reload the page. jQuery helps streamline tasks like event handling, making the platform more efficient. During development, XAMPP is used to run the platform locally on a localhost, mimicking how it will function on a live server. The platform also meets GDPR privacy standards to protect user data. Altogether, these technologies ensure the platform is safe, interactive, and provides an engaging experience for students.

Content Analysis with Competitor Websites

I learned a lot about the features and capabilities that could improve the Student Learning Adventure Platform by examining these systems. I gave priority to features like peer-to-peer Q&A, real-time updates, and content moderation, taking inspiration from sites like Piazza and Stack Overflow. My focus on developing a modular structure that could grow with future features like course management and user interaction tools was informed by the adaptability and scalability of Moodle and Edmodo. Last but not least, Google Classroom's intuitive, user-

friendly interface was influenced by its user-friendly design and integration of external tools. The creation of a platform that prioritizes student cooperation, usability, security, and scalability—with potential for future growth into more sophisticated features—was inspired by these ideas taken together.

Below are screenshots of the websites showing where some deep analysis came from



SEO Issues and Recommendations

A number of SEO factors were taken into account when developing the Student Learning Adventure Platform in order to improve search engine performance, visibility, and usability. The platform concentrated on fixing typical problems like slow page loads by optimizing scripts and pictures, making sure that the design was flexible for mobile, and making SEO-friendly URLs for ranking and clarity. For features like Q&A, structured data (schema markup) was taken into consideration to increase search engine comprehension and increase the possibility of rich snippets.

SEO Analysis and Integration in the Student Learning Adventure Platform

Essential SEO techniques are incorporated into the Student Learning Adventure Platform to improve user engagement and exposure. Relevance and click-through rates were increased by optimizing meta tags for each page with distinct titles and descriptions. To improve accessibility and boost image search ranks, alt text for photos was added. In order to facilitate user navigation and search engine crawling, internal linking was purposefully created to link related material. The development of content suited to students' needs was directed by keyword analysis, guaranteeing that it was pertinent to the platform's user base.

Features like responsive design for mobile usability and quick page loading were given priority in order to further maximize visibility. Analytics capabilities aid in tracking

performance and directing continuous enhancements, while social sharing options were incorporated to promote the spread of content. The platform itself was immediately integrated with these SEO tactics, guaranteeing best practices for search engine optimization hence it is accessible, functional and discoverable for students.

Legal and Ethical Issues and ethical considerations

Legal and ethical considerations, like protecting user data, ensuring accessibility, and respecting copyright, are key to building a responsible platform. The Student Learning Adventure Platform follows GDPR rules to keep user data safe, meets accessibility standards so everyone can use it easily, and uses properly licensed content to avoid copyright issues. It also focuses on transparency and fairness, creating a space where students feel respected and supported. These efforts ensure the platform is not only compliant with legal standards but also ethical in meeting the needs of its users.

Discussion on GDPR and accessibility standards.

The Student Learning Adventure Platform places a high priority on data safety and accessibility to guarantee inclusivity and security. The platform protects user data by complying with GDPR by using transparent permission procedures and safe storage. Usability for everyone, including those with impairments, is ensured by accessibility features including user-friendly navigation and support for assistive devices. These actions support the project's objectives and give a safe and welcoming environment to a wide range of users.

Compliance with Usability, Accessibility features, and Web Standards

The design of the Student Learning Adventure Platform placed a high priority on web standards, usability, and accessibility. While accessibility features like keyboard navigation and screen reader compatibility guarantee that all users, including those with impairments, can interact with the platform efficiently, the user interface is simple enough for administrators and students to navigate. The platform works effectively on a variety of browsers and devices since it complies with web standards including HTML5 and CSS3. The database's use of MySQL facilitates the effective administration of academic courses and user data. This strategy guarantees that the platform is inclusive, easy to use, and has strong technical foundations.

System Design and Development Approach

The Student Learning Adventure Platform was developed using a methodical and planned process that made use of contemporary online tools and technology to produce an engaging and user-friendly platform. The procedure was divided into multiple crucial stages including the mentioned below:

1. Planning and Requirements Stage

Initially, the main objective was to create a forum where students could publish, respond to, and engage with academic questions pertaining to their modules or courses. With distinct responsibilities for administrators and students, the system needed to be user-friendly, scalable, and safe. This involved configuring data management, user authentication, and interaction tools including question and post viewing.

2. System Design stage

During this stage, the platform's design was outlined using wireframes and flowcharts, which made sure that every page was simple to use and that the interactions made sense. Both the admin-side (content management, user management) and student-side (question posting, question interactivity) layouts were shown in the wireframes. The user journeys were mapped out with the aid of flowcharts, which illustrated how users would move through the system (e.g., login, post question, admin review).

3. Database design stage

To hold user data, queries, responses, and other materials, a MySQL database was created. Relationships between the tables for users, user_profiles, modules, notifications, answers, posts, replies, modules, contact_messages and comments were specified in the database structure. These tables served different functionalities based on how they are named. The usage of foreign keys to connect posts to users and modules was a crucial component of the architecture. This was enabled using localhost/phpMyAdmin.

4. Back-End Development

The back-end was primarily built using PHP and MySQL. The PHP files created including; login.php, register.php, index.php, create_post.php, edit_post.php, delete_post.php amongst other php files were linked to the database via the central config.php file, ensuring smooth interactions with the MySQL database. The back-end included:

User Authentication: For secure login and session management.

Post Management: Allow students to create, edit, delete and view answers to their questions.

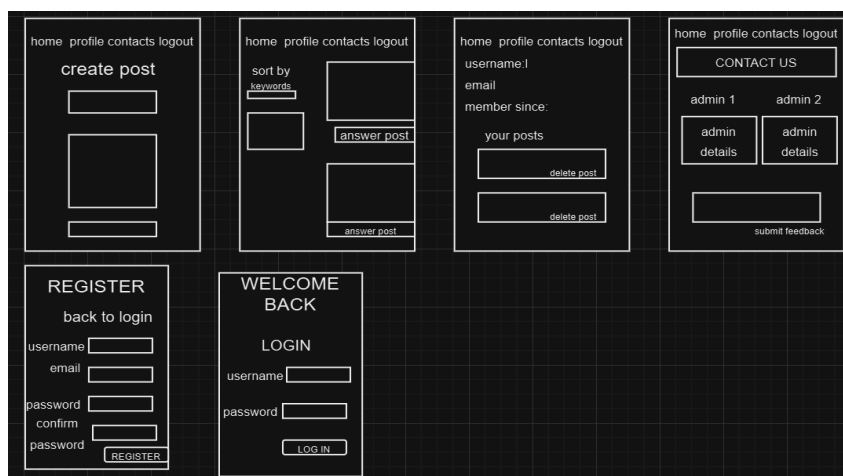
Admin Panel (admin dashboard): Enabling administrators to manage users and posts, ensuring content quality and relevance whilst also allowing users to contact admins if they need help.

5. Front-end development

The front-end was designed to be user-friendly and responsive, ensuring it worked well. HTML, CSS, and JavaScript, were used to create a clean and intuitive layout. Students are enabled to upload images with their posts, making it easier to share relevant visuals along with their questions and also a user is allowed edit his posts and answer other users' posts. This combination of features aimed to improve usability and engagement on the platform.

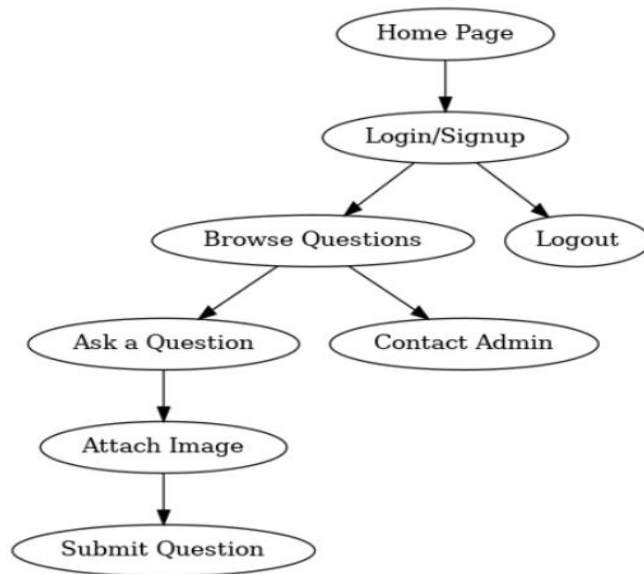
Wireframes

The image below shows the wireframes for the website

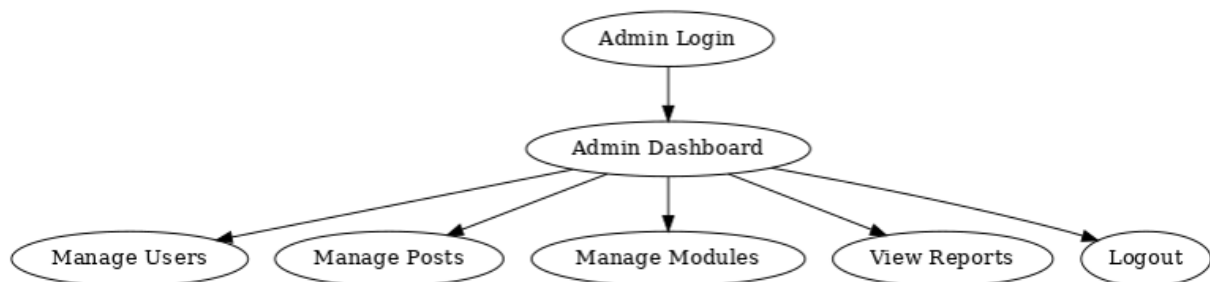


Flowcharts

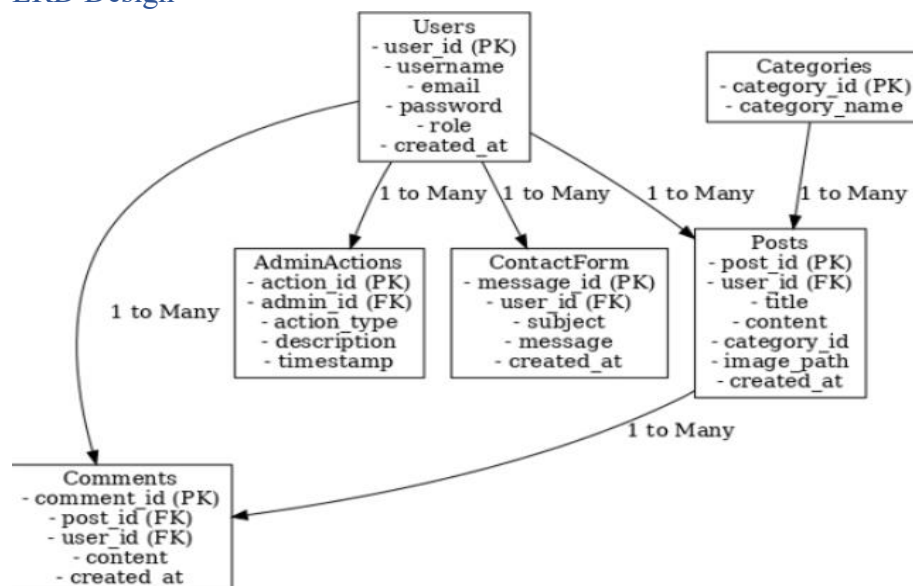
A. Client-side flowchart



B. Admin-side flowchart



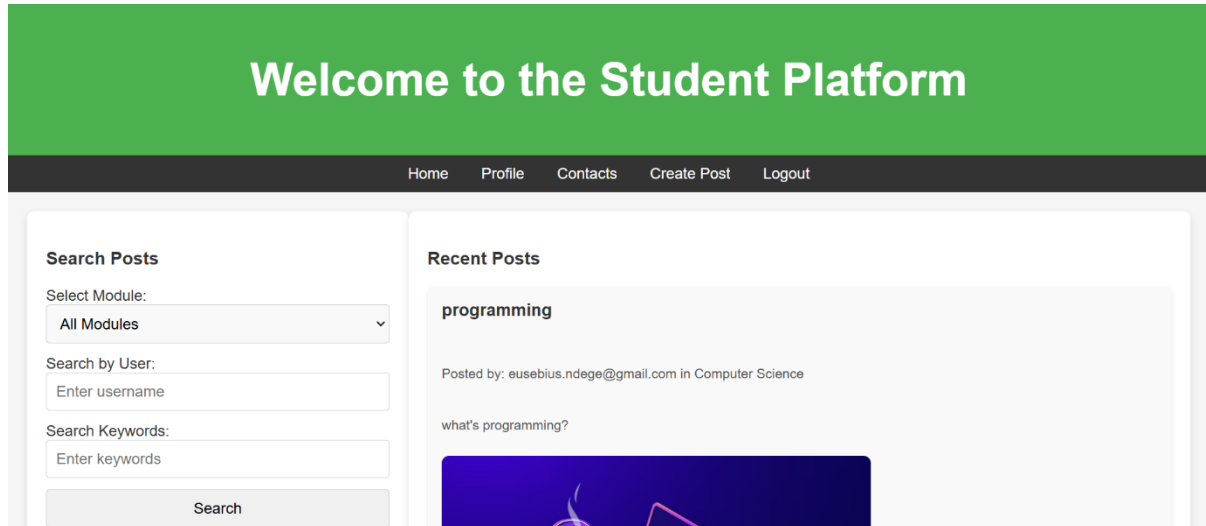
ERD Design



System Implementation

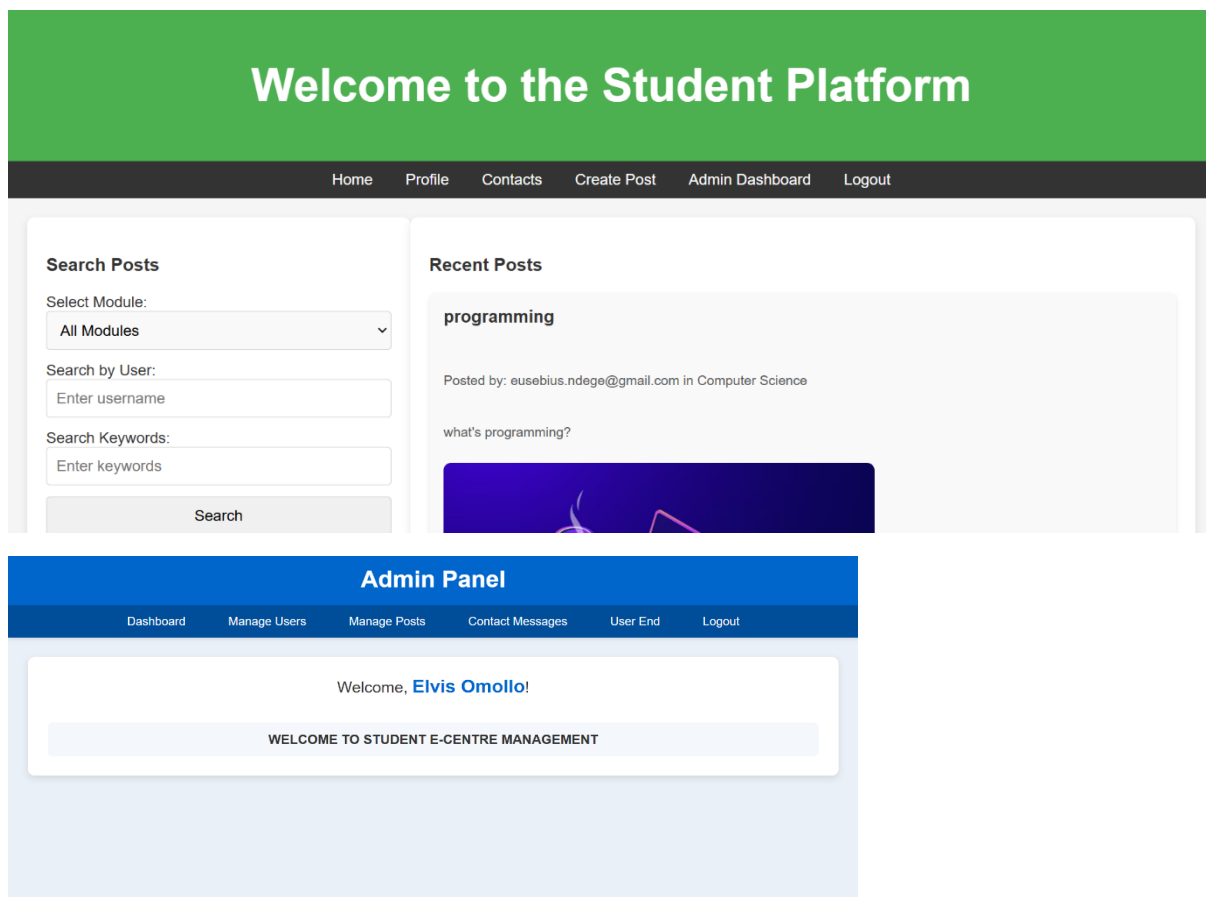
A. Client-side Functionality

It is visible that client has limited functionality revolved around seeing posts on the home page, access his profile, contact admin, create a post, answer it, update the post and also delete the post.



B. Admin-side Functionality

The admin end as shown below has more navigation buttons enabling the admin to manage users, manage posts, access contact messages from users and also access the user end by clicking the admin dashboard.



6. Integration and Testing stage

Following their construction, the separate parts were incorporated into the platform. Thorough testing was done to make sure the system functioned properly throughout. Performance testing was done to make sure the platform could manage an increasing number of users and posts, and usability testing was done to make sure users could easily navigate the platform.

System Testing

Test cases and results (validation, functionality testing).

The testing process for the Student Learning Adventure Platform involved a comprehensive suite of test cases designed to validate the platform's functionality, user experience, and database performance. The tests were carried out step-by-step, each focused on a specific aspect of the platform, ensuring everything functioned as intended.

1. **User Registration & Login:** Tests were performed to verify that new users could register successfully and log in with their credentials. These tests confirmed that the registration and authentication processes worked as expected.
2. **Post Creation & Image Upload:** The platform's post creation feature, including the ability to upload images, was tested. This ensured that students could create posts with text and images and files, which would be displayed in the forum.
3. **Admin Panel Functionality:** The admin's ability to manage posts and users was tested. Admins were able to edit, delete posts, and manage user accounts on their admin dashboard confirming that administrative controls were functioning properly. They admins are also enabled to access a user's end too. The was enables through the adm.php file.
4. **Search & Navigation:** The search functionality was tested to verify that posts containing relevant keywords could be found. Navigation through the platform was also tested for smooth usability.
5. **GDPR Compliance & Data Privacy:** The platform's GDPR compliance was validated by checking for data privacy notices and options for data deletion, ensuring user privacy was respected.
6. **Responsive Design:** Tests were carried out on multiple devices and different browsers to ensure the platform's responsive design adjusted to different screen sizes.
7. **Database Connectivity:** Tests in localhost myphpAdmin ensured that all data entered through the platform was stored correctly in the MySQL database and could be retrieved from different tables as needed
8. **Overall Performance & Security:** Functional and security tests were conducted to ensure that the system handled user interactions securely and efficiently.

The tests were executed on a local development environment and various browsers such as chrome and Microsoft edge and devices used to simulate real-world user experiences. Each test case was checked thoroughly to ensure that all expected outcomes were met and the

platform operated seamlessly. Based on the successful results of these tests, the platform was considered ready for deployment.

NOTE: Proof of these tests has been clearly Outlined in the appendices with screenshots showing the tests

7. Final touches

The platform was enhanced with SEO optimizations, including meta tags, alt text for images, and internal linking of the php files. It was also designed with GDPR compliance in mind, ensuring that user data was protected and handled securely through the admin panel whereby the admin can monitor the users and regulate activity.

Conclusions and Future Recommendations

The Student Learning Adventure Platform was effectively created to give students a place to interact, exchange information, and pose inquiries. It included essential elements including responsive design, real-time updates, and administrative controls, which improved the platform's usability and functionality. The platform satisfies important needs like GDPR compliance and is safe, scalable, and easily accessible. It also adheres to best standards for web development.

The platform facilitates academic collaboration by giving administrators and students the means to control information and communicate. Dynamic interactions and seamless performance were guaranteed by the integration of PHP, MySQL, and JavaScript. Its usability and functionality were validated through testing, demonstrating its value as a tool for increasing student engagement.

In the future, the platform might use some significant enhancements. Multi-factor authentication would improve security, and students would find the experience more interesting if it were gamified with leaderboards or prizes. Students could more readily access the platform while on the go with the aid of a mobile app, and by including analytics, we could gain a deeper understanding of user activity. Real-time assistance could be offered by an AI chatbot, and the platform's ability to accommodate expansion will be guaranteed by optimizing it for scalability. Lastly, incorporating community tools like discussion boards and private messaging could encourage deeper student connections.

Reflection on project outcomes

Looking back on the creation of the student learning platform, I can say that it was a satisfying and demanding effort. Aligning the design with the user experience, integrating a scalable and secure database, and guaranteeing accessibility compliance were among the main issues. I resolved design problems by making changes to wireframes and flowcharts, and PHP PDO took care of security issues. Usability was improved via AJAX real-time updates, picture uploads, and an easy-to-use interface. Notwithstanding the difficulties, the project offered insightful information about platform expansion, user experience, and web development.

References

Harvard-referenced sources and citations.

Baeza-Yates, R. and Ribeiro-Neto, B. (2011) Modern Information Retrieval: The Concepts and Technology behind Search. 2nd edn. New York: Addison-Wesley.

This book provided foundational knowledge about designing web-based platforms and optimizing them for search.

Nielsen, J. (1993) Usability Engineering. San Francisco: Morgan Kaufmann.

This text informed the platform's usability and accessibility considerations.

Welling, L. and Thomson, L. (2017) PHP and MySQL Web Development. 5th edn.

Indianapolis: Addison-Wesley Professional.

This source guided the integration of PHP PDO and MySQL database functionalities.

Robbins, J. N. (2018) Learning Web Design: A Beginner's Guide to HTML, CSS, JavaScript, and Web Graphics. 5th edn. Sebastopol: O'Reilly Media.

The book helped in creating responsive and interactive front-end designs.

ISO (2018) ISO 9241-11:2018 - Ergonomics of human-system interaction – Usability: Definitions and concepts. Available at: <https://www.iso.org> (Accessed: 1 December 2024).

Referenced to ensure compliance with international usability and accessibility standards.

European Union (2018) General Data Protection Regulation (GDPR). Available at:

<https://gdpr-info.eu/> (Accessed: 1 December 2024).

Used to ensure the platform adheres to privacy and data protection standards.

Duckett, J. (2014) HTML and CSS: Design and Build Websites. Indianapolis: Wiley.

This book supported the creation of clean and structured front-end elements.

Appendices

Annotated Walkthrough

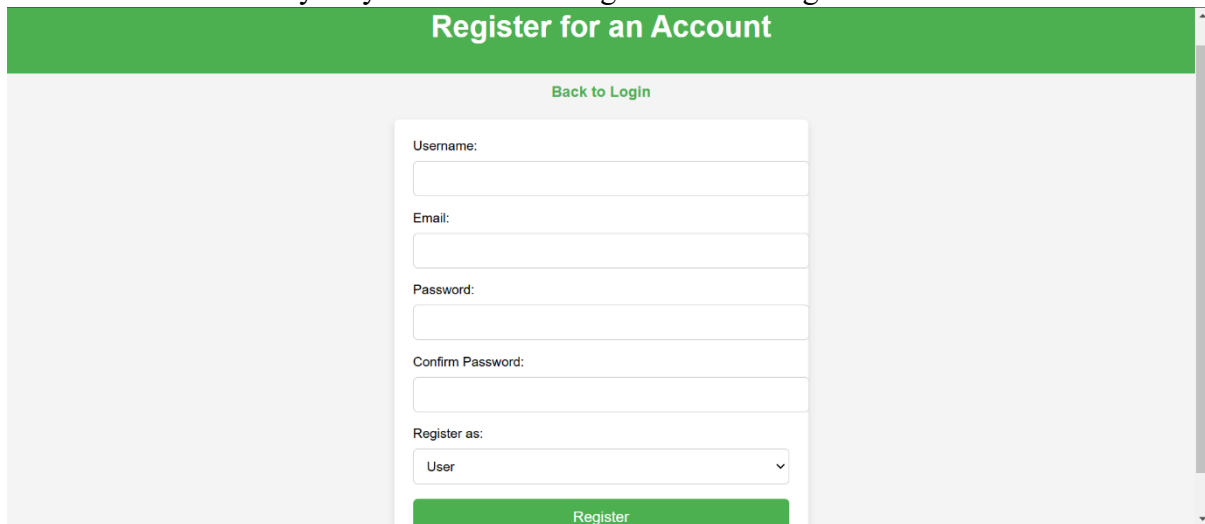
Supporting materials, including detailed testing evidence and additional screenshots.

Screenshots and descriptions showcasing system functionality.

A. USER VIEW

1. Register page (register.php page)

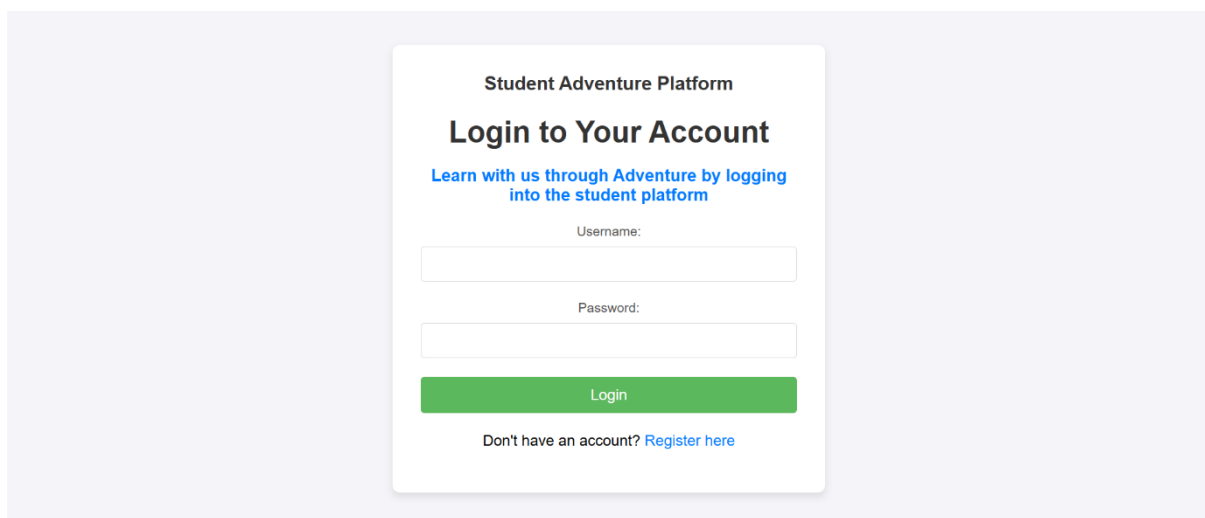
Users are enabled to create accounts they can use to login with through this page and stored in the database whereby they are allowed to login after creating an account.



The screenshot shows a web page titled "Register for an Account" with a green header. Below the header is a green button labeled "Back to Login". The main content area is a light gray box containing a registration form. The form has the following fields: "Username:" with a text input, "Email:" with a text input, "Password:" with a text input, "Confirm Password:" with a text input, and "Register as:" with a dropdown menu showing "User". At the bottom of the form is a green button labeled "Register".

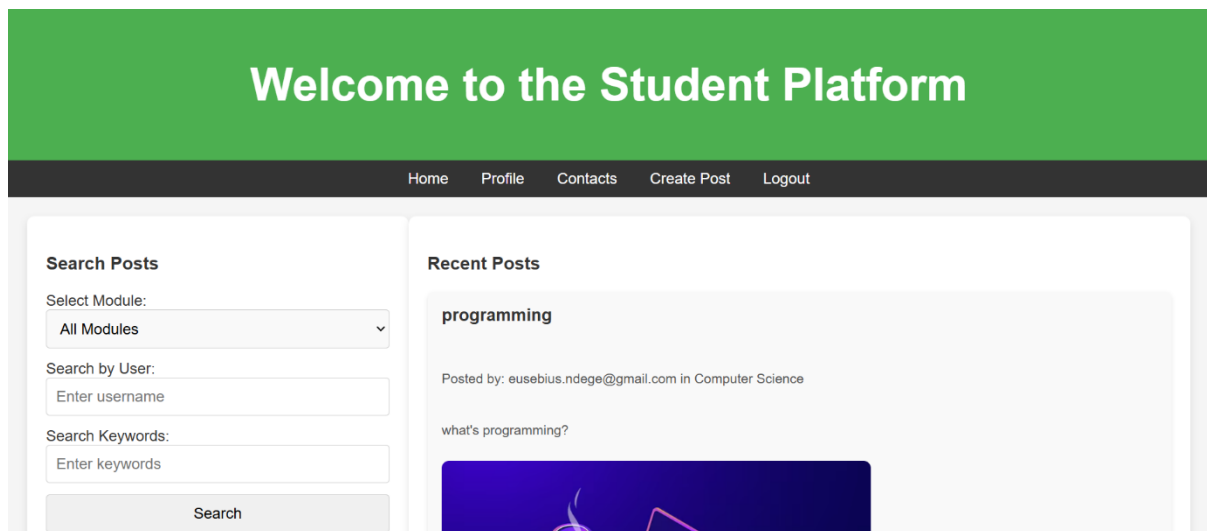
2. Login page (login.php)

It enables the user to login after referring back to the database through the config.php page, login.php and admin.php page

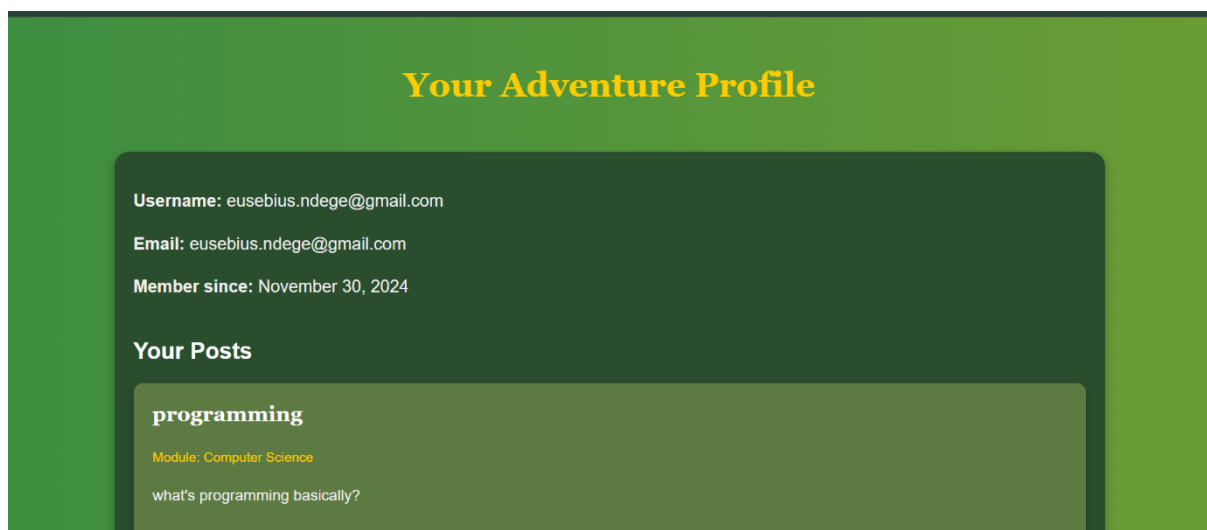


The screenshot shows a web page titled "Student Adventure Platform" with a subtitle "Login to Your Account". Below the subtitle is a blue link that says "Learn with us through Adventure by logging into the student platform". The main content area is a light gray box containing a login form. The form has the following fields: "Username:" with a text input, and "Password:" with a text input. At the bottom of the form is a green button labeled "Login". Below the button is a link that says "Don't have an account? Register here".

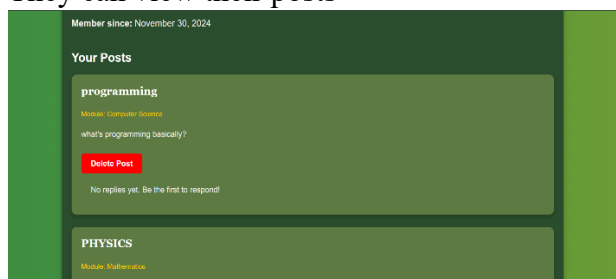
3. User Dashboard ~ Home page ~ (index.php)



4. Profile page (profile.php) ~ Showing the username, the date the user joined and the view their posts and answers (delete posts)

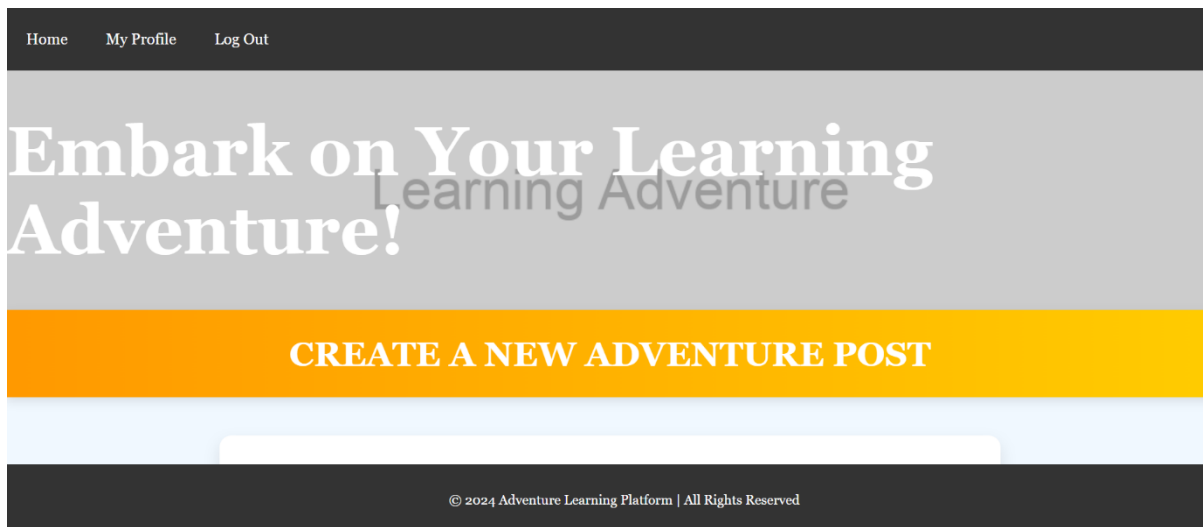


They can view their posts



5. Creating post page (create_post.php)

Enables creation of a post which enables attachment of an image alongside it and then uploaded on the home page through linking up with the index.php

This screenshot displays a form for creating a new post. The form is set against a light blue background. It includes a 'Post Title:' label followed by a text input field with the placeholder 'Enter your learning adventure title'. Below this is a 'Post Content:' label followed by a large text area with the placeholder 'Describe your adventure or learning experience...'. A 'Select Module:' label is followed by a dropdown menu currently showing 'Computer Science'. There is a green button labeled 'Upload Image (optional):'. Below the button is a file upload area with a 'Choose file' button and the text 'No file chosen'. At the bottom of the form is an orange button labeled 'Post Your Adventure'.

6. Contact admin page (contacts.php)

Enables users to contact the admins and feedback is provided back

Contact Us

Home Profile Create Post Logout

John Doe
Administrator
[Contact John Doe](#)

Jane Smith
Support Agent
[Contact Jane Smith](#)

Administrator
[Contact John Doe](#)

Support Agent
[Contact Jane Smith](#)

Contact John Doe

Write your message...

Send Message

7. Logout (logout.php)

After logging out the user is directed to the login page(login.php) by the logout.php file.

Student Adventure Platform

Login to Your Account

Learn with us through Adventure by logging into the student platform

Username:

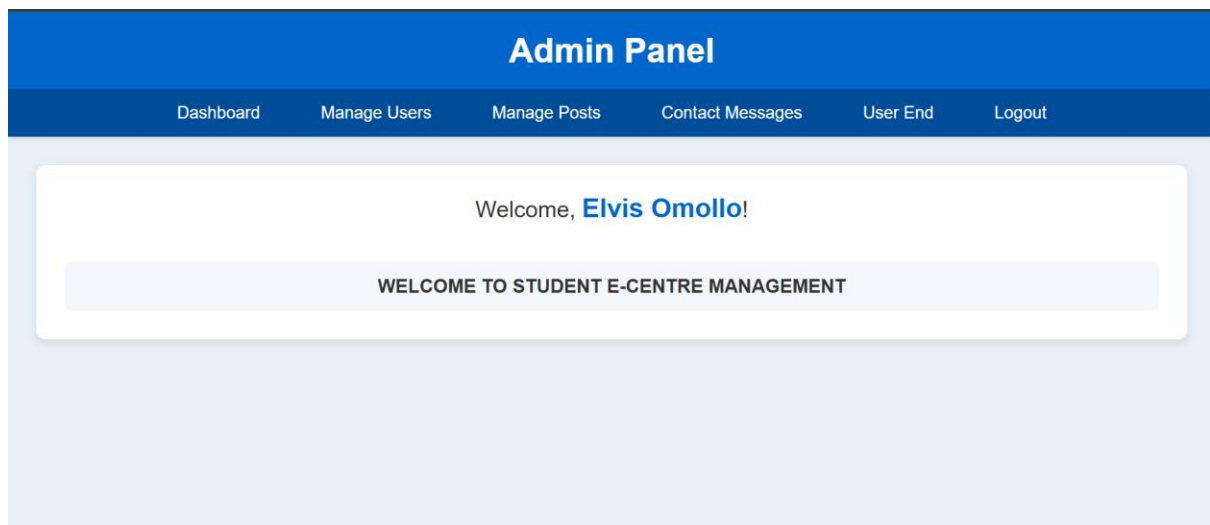
Password:

Login

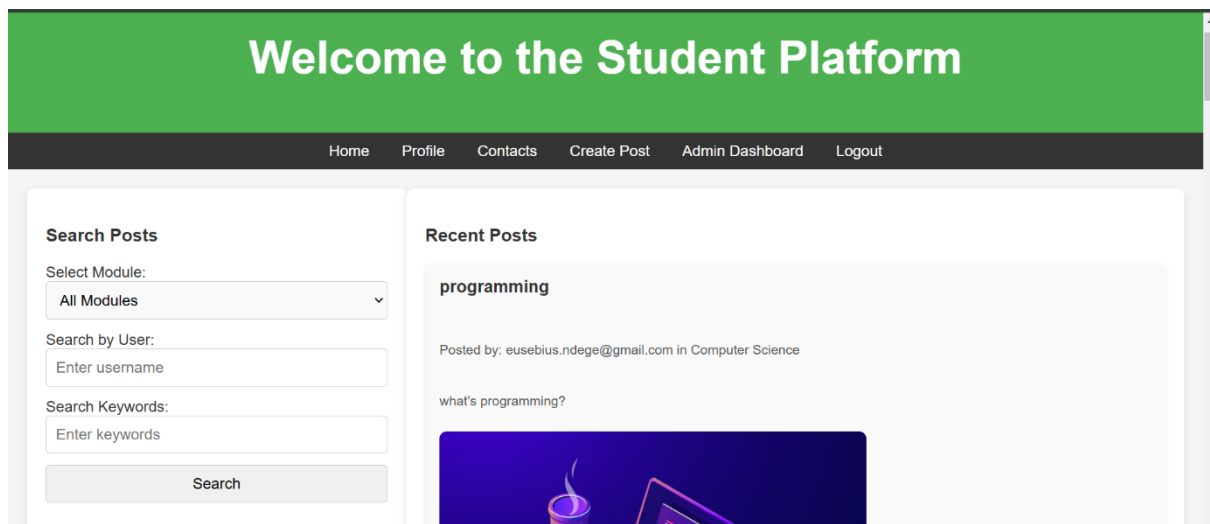
Don't have an account? [Register here](#)

B. ADMIN VIEW

1. Admin dashboard

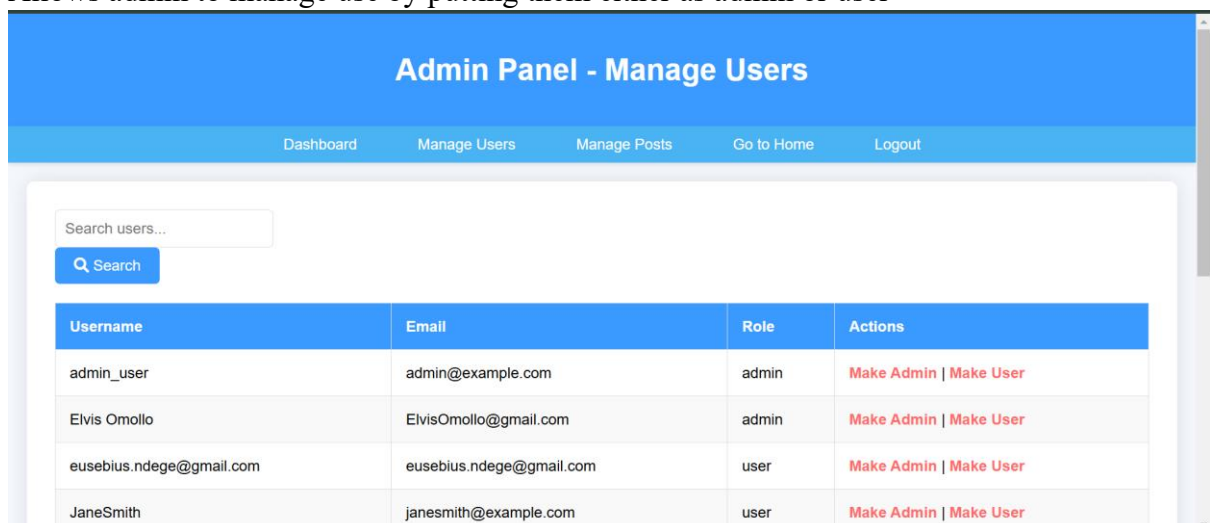


2. Admin home page(add_admin.php and admin.php)

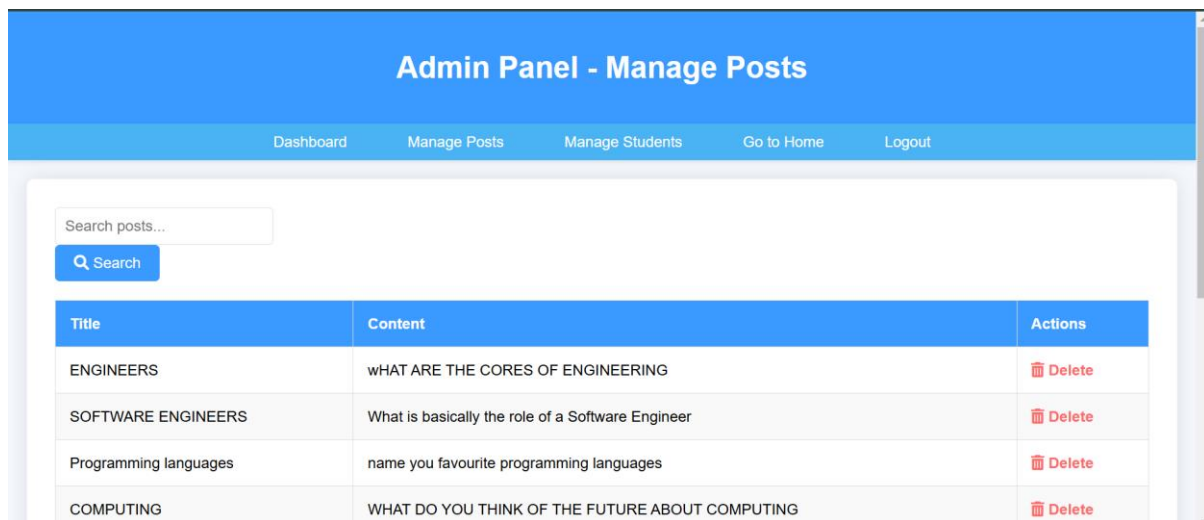


3. Manage users (manage_users.php)

Allows admin to manage use by putting them either as admin or user



4. Manage posts(manage_posts.php) – Enables management of posts by deleting them.



5. User end of the admin (user_end.php)

Goes back to home page(index.php) with an addition admin dashboard navigation button

